A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 31

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1** Negative **1** Positive **1** Negative **1** Positive **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** The response to a change acts to re-establish / return the system to the previous equilibrium **1** So the change does not become self-sustaining unless a large unusual event overwhelms the negative feedback **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** eg the possible release of methane hydrate at the end of the Permian era **1** Overwhelmed the negative feedback mechanisms, causing climate change and a mass extinction / a new equilibrium **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Soil degradation, reduced water availability and loss of natural pest predators would reduce yields (negative feedback curtailing the activity) **1** Instead of reducing the activity, farmers use fertilisers, irrigation and pesticides, which further damage the natural systems that support food production **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 8 nights **1** The running mean fluctuates widely with few samples but stabilises at 21-22 from 8 nights onwards – extra samples do not change it significantly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Moth activity varies with temperature, wind, rain and season (adults only fly for part of the year) so catches vary independently of population size **1** Sampling at different conditions / times would make results not comparable **1** Timing between samples should be chosen to detect the temporal variation of interest (eg weekly through the season) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1 Accuracy:** how close a recorded result is to the real value **1 Precision:** the smallest interval between possible recorded values (eg a ruler marked in mm is more precise than one in cm) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value • **precision** → readings close together / fine resolution	2
02.4	**1 mark each** 1 mark for each of the 3 points below. **1** A null hypothesis (eg no difference / no correlation) can be tested statistically and accepted or rejected at a stated confidence level **1** Standardised techniques ensure results collected at different times, places or by different people are comparable **1** Other variables are controlled or monitored so their effect can be assessed – allows the study to be repeated / checked (reliability) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **null hypothesis** → a statement that there is no significant difference or correlation • **correlation** → a relationship in which the two variables change together • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 5 points below. **1 Timber:** building construction, furniture, tools, telegraph poles, concrete shuttering (oak, beech, mahogany, teak, conifer softwood) **1 Fuel:** fuelwood and charcoal - still the main energy source for most people in LEDCs **1 Food:** fruit, nuts, bushmeat, livestock fodder – crops of forest origin (coffee, cacao, bananas, Brazil nuts) – pigs and chickens were forest animals **1 Fibres:** paper (wood pulp cellulose), cotton, viscose / rayon **1 Medicines:** quinine (cinchona), taxol (yew), aspirin (willow) – most species not yet researched **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** Wood is mainly cellulose which is hard to digest, so carbon stored in wood has a long residence time – total forest carbon (vegetation plus soil) is about double that in the atmosphere **1** Growing trees sequester CO_2 by photosynthesis and release oxygen – forest soils hold more carbon than the trees (especially boreal) **1** Planting / protecting forests removes CO_2 and reduces the enhanced greenhouse effect **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime • **enhanced greenhouse effect** → extra warming caused by added greenhouse gases **Do not credit** • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Highest transpiration of any ecosystem increases downwind rainfall – interception increases evaporation – deep organic soils retain water and delay runoff so river flow is more even and flooding reduced **1** Low albedo absorbs sunlight – heat stored and released at night reduces day-night temperature extremes – canopy microclimate: lower light, wind, higher humidity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-364; AQA spec 3.5.3.1.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Irrigation depletes rivers and aquifers (eg Aral Sea, Ogallala) – increases evaporation **1** Soil erosion and compaction reduce water retention and infiltration so runoff and river flow fluctuations increase **1 Evapotranspiration changes:** increased in irrigated arid areas, decreased where forests were cleared **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface • **Evapotranspiration** → evaporation plus transpiration **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	3
04.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Seasonal / local:** avoids energy for heating, lighting greenhouses out of season and long-distance transport (food miles) and the pollution this causes **2 Organic:** uses natural processes (rotation, legumes, manure, biological control, predator habitats) instead of artificial fertilisers and pesticides, so less pollution, energy use and harm to non-target species – free-range and Fairtrade also accepted for welfare / social reasons **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** Food shortages after 1945 (dependence on US food aid) – farmers could not afford loans to invest so grants funded machinery, drainage, hedgerow removal, liming, improved livestock **1 Guaranteed market / price support:** government bought surpluses to hold prices up (and sold stores in poor years) giving farmers security to increase output **1 Consequences:** loss of hedgerows and wetlands, intensification, pesticide and fertiliser pollution – later food surpluses (butter mountains) that could not be sold without harming LEDC producers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-333; AQA spec 3.5.1.3-3.5.1.4.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - radio waves do not travel through seawater – acoustic or satellite tags transmit position	1
05.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2** Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved **2** Radio / GPS tags or collars – leg rings / wing tags on birds – camera traps – DNA from droppings to confirm identity and diet **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 600 + (90 + 150) - (220 + 80) = 540 **1** Stock fell by 60 so the catch exceeds the MSY / unsustainable **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (thousand tonnes) is already printed on the answer line. • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks. • A bare wrong answer with no working.	2
06.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** A + G - M = 90 + 150 - 80 = 160 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (thousand tonnes) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** Catch is landed and weighed at ports so it is recorded **1** Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed **1** Representative sampling of a large enough proportion of the area is impractical / expensive – survival of eggs and larvae varies greatly between years **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
06.4	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 CPUE:** if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained – a better measure than catch size which technology can prop up **2** Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2.	
07.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2 Species:** a group of closely related organisms that naturally interbreed to produce fertile offspring and are reproductively isolated from other groups (Passer domesticus, house sparrow) – the only 'real' taxon **2 Niche:** the role a species plays in its habitat - how it uses resources and relates to other species (tawny owl: nocturnal carnivore nesting in tree holes eating small mammals) – two species cannot share exactly the same niche in one habitat **2 Community:** all the populations of all species (plants, animals, fungi, bacteria, protists, archaea) living in an area **2 Biome:** a large geographical region with a specific climate and characteristic community, including all areas where it occurs (all tropical rainforests, tundra, savannah) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **Niche** → the role a species plays and the conditions it needs	8
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** An ecosystem is the biotic community plus abiotic factors of one area, relatively self-contained (a particular coral reef) **1** A biome includes all separate ecosystems with the same climate and community type (all coral reefs) – each reef is a different ecosystem within the same biome **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biotic** → living factors / effects of other organisms • **abiotic** → non-living / physical factors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 82-85; AQA spec 3.1.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** Mars has low mass so weak gravity and could not retain a dense atmosphere **1** Very low atmospheric pressure means water would boil / evaporate away even at low temperatures **1** Greater distance from the Sun gives low insolation and a mean temperature of -60 °C so water is frozen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	3
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Mass great enough for gravity to retain an atmosphere containing the elements needed (C, H, O, N as CO2, methane, ammonia, water vapour) and to maintain pressure so liquid water does not boil **1** Distance from the Sun (the 'Goldilocks zone') gives insolation producing temperatures at which water is liquid (0-35 °C over most of the surface) and proteins are not denatured **1** Too close (Venus) - too hot – too far (Mars) - too cold **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	3
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** The molten core produces a magnetosphere that deflects the solar wind / charged particles, preventing biologically damaging radiation reaching the surface **1** The 24-hour rotation limits heating by day and cooling by night, reducing temperature extremes (Venus' slow rotation gives extreme conditions) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** The tilted axis causes seasonal variation in insolation as the Earth orbits the Sun **1** Producing seasons (temperature, day length) to which species' life cycles (breeding, migration, dormancy) are adapted – the Moon's gravity keeps the tilt stable and creates tides **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** The nucleus of a stem cell from the endangered species is transferred into an empty egg of a closely related species (southern white rhino) **1** The embryo is implanted into a surrogate female of the related species which gives birth to a northern white rhino **1** Very low success rates with wildlife so far – deep-frozen cells could allow clones even after the last individuals die **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
09.2	**1 mark each** 1 mark for each of the 3 points below. **1 Micro-propagation:** tissue culture producing many cell clusters from one plant, each grown into a genetically identical plant **1 Black poplar:** cuttings / pollarded stems of pure native trees are rooted to produce many identical trees **1** Needed because the flowers are pollinated by introduced subspecies producing hybrids, so cloning maintains the pure gene pool (but no genetic diversity) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
09.3	**1 mark each** 1 mark for each of the 4 points below. **1** Stores seeds of wild plants (especially threatened species and crop wild relatives) so species are not lost if extinct in the wild **1** Seeds collected worldwide – target of 10 000 seeds per species to maintain a diverse gene pool **1** Stored dry and refrigerated / frozen in an underground reinforced vault to reduce risk of accident **1** Seeds germinated periodically to check viability and replenish stocks – can be used for reintroduction / breeding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 35-37; AQA spec 3.1.2.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Protected areas:** Korup NP Cameroon (1260 km^2, 600 tree species, endemic Ancistrocladus korupensis with anti-HIV / antimalarial alkaloids) – Tumucumaque NP Brazil (39 000 km^2, adjoining French Guiana reserve) – Alto Maues – Rio Bravo Belize funded by charities, companies and ecotourism • Debt for nature swaps – sustainable exploitation: selective logging of high-value timber (mahogany) funding conservation – ITTO aims (largely unmet) – CITES listing of timber species (Honduras mahogany Appendix II) • Ecotourism income (but visitor damage in Costa Rica, Malaysia) – community projects and employment – carbon sequestration payments • **Effectiveness:** protected areas work where enforced but face illegal logging, mining, poaching, roads, conflict – drivers (population growth, demand for beef, soya, palm oil, timber, minerals) originate outside the forest – climate change cannot be controlled locally • **Judgement:** combination of large protected areas, economic alternatives for local people and reduced international demand needed – overall loss continues **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **endemic** → found naturally only in that one place • **sustainable** → meeting present needs without stopping future generations meeting theirs • **CITES** → the treaty that controls international trade in endangered species	9
	Do not credit • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'CITES bans all trade in wildlife' - it controls and licenses trade, with different rules for each appendix	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Protected areas are the core of rainforest conservation. Korup National Park in Cameroon protects 1260 km^2 with about 600 tree species, including the endemic *Ancistrocladus korupensis* whose alkaloids have anti-HIV and antimalarial activity; Brazil's Tumucumaque National Park covers 39 000 km^2 and adjoins a reserve in French Guiana; the Rio Bravo area in Belize is funded by charities, companies, ecotourism and selective logging of high-value mahogany that does not damage the forest long term. Debt-for-nature swaps have cancelled Guatemalan and Brazilian debt in return for protection, CITES listing restricts trade in Honduras mahogany, the ITTO sets sustainable timber aims, ecotourism and community employment give local people an income from standing forest, and carbon sequestration payments are emerging.	
	Where they are enforced, protected areas work, but they face illegal logging, mining, poaching, road building and conflict, and ecotourism itself has damaged sites in Costa Rica and Malaysia. The deeper problem is that the drivers of deforestation - population growth and international demand for beef, soya, palm oil, timber and minerals - originate outside the forest, and climate change threatens even well-protected areas; the ITTO's aims are largely unmet and overall loss continues.	
	The most effective schemes are those that combine large protected areas with economic alternatives for local people, such as Rio Bravo, but they can only slow the loss unless demand in importing countries also falls.	
	Weaker answer - Level 2 (4 marks)	
	Rainforests are protected by national parks where logging is banned, for example in Brazil. Debt-for-nature swaps mean a country's debt is cancelled if it protects its forest. Ecotourism brings money from tourists so the forest is worth more standing than cut down. Sustainable logging only takes some trees. These efforts have saved some areas but deforestation is still happening because of illegal logging and farming, so they have only been partly effective.	
	What makes the difference	
	• Level 2 covers the main types of effort and a fair conclusion, but the question says 'describe' - Level 3 gives named examples with figures (Korup, Tumucumaque, Rio Bravo) as the mark scheme expects.	
	• Level 3 explains *why* effectiveness is limited (drivers originate outside the forest, ITTO aims unmet, climate change) rather than just noting continued loss.	
	• The judgement in Level 3 identifies what works best and what would be needed; Level 2 stops at 'partly effective'.	
10.2	1 mark for the point below. **1** Carbon sequestration reducing climate change – genetic resources / medicines – rainfall regulation – global biodiversity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 56-57; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Species selection:** market, range of tolerance, controllability • **Abiotic control:** temperature, light, water, soil fertility / nutrients, aeration, salinity, pH, CO_2, topography, relief, wind - methods and examples • **Biotic control:** pest groups, cultural control, biological control (successes and failures), sterile males, pheromones, genetic resistance, pesticides (properties), antibiotics, pollinators, soil biota • **Manipulation of food species:** population density, monocultures, selective breeding, asexual reproduction, cross-breeding, AI / embryo transfer, GM – energetics - intensive vs extensive, energy subsidies, ratios, FCR, food chain losses • **Impacts:** habitat loss / change, reduced biodiversity, introduced species, pesticide and nutrient pollution, GHGs, hydrological changes, soil erosion – positive: plagioclimax habitats • Social, economic and political influences shaping methods • Sustainable strategies and evaluation	25
	– conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-337; AQA spec 3.5.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Physical features:** mass / gravity retaining an atmosphere and giving pressure for liquid water – distance from the Sun and the Goldilocks zone – speed of rotation reducing temperature extremes – tilt producing seasons – magnetic field deflecting the solar wind – the Moon stabilising the orbit and producing tides • **Importance of liquid water:** solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer • **Development of life:** Archaea at vents – photosynthesis – oxygen reacting with iron then accumulating – ozone layer allowing colonisation of land – aerobic organisms • **Life changing conditions:** carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases	25
	• **Methods of studying past conditions:** proxy data, isotope ratios, ice cores, Mg:Ca ratios – understanding of continental drift and ocean currents • Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	

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